

PRACTICE SHEET

EXERCISE-I

(Adsorption)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. $\frac{x}{m} = KC^{1/n}$ where C = Concentration of a solution, x = wt of adsorbed solute, m = Wt of adsorbent.
The equation represents
 - 1) Langmuir adsorption isotherm
 - 2) Freundlich adsorption isotherm
 - 3) Arrhenius equation
 - 4) Chemisorption isotherm
2. A graph drawn between ' $\log \frac{x}{m}$ ' and $\log p$ on 'Y' and 'X' axis respectively could be ____ regarding physical adsorption
 - 1) A hyperbola
 - 2) A parabola
 - 3) A straight line with positive slope
 - 4) A straight line with negative slope
3. Which of the following cannot act as an adsorbent?
 - 1) Silica gel
 - 2) Clay
 - 3) Oxygen gas
 - 4) Activated charcoal
4. Which can adsorb maximum amount of H_2 ?
 - 1) A platinum black
 - 2) Powdered palladium
 - 3) A platinum rod
 - 4) Nickel sphere
5. During activation of charcoal ____
 - 1) O_2 is adsorbed on charcoal
 - 2) Moisture is absorbed on charcoal
 - 3) Pre adsorbed material is desorbed
 - 4) Charcoal is covered with inert gas
6. The energy of molecules present on the surface of a substance is ____
 - 1) Equal to that of molecules in bulk
 - 2) Greater than that of interior molecules
 - 3) Less than that of interior molecules
 - 4) Dependent on nature of substance
7. The Langmuir adsorption isotherm is given by
 - 1) $\frac{x}{m} = kP^{1/n}$
 - 2) $\frac{x}{m} = \frac{aP}{1+bP}$
 - 3) $\frac{x}{m} = \frac{bP}{a}$
 - 4) $\frac{x}{m} = kP^n$
8. At very high pressure the Langmuir adsorption isotherm takes the form of
 - 1) $\frac{x}{m} = kP$
 - 2) $\frac{x}{m} = \frac{a}{b}$
 - 3) $\frac{x}{m} = \frac{1}{1+aP}$
 - 4) $\frac{x}{m} = P$
9. Based on Langmuir adsorption isotherm, the intercept in the graph ($\frac{m}{x}$ versus $\frac{1}{p}$) is equal to
 - 1) $\frac{1}{a}$
 - 2) $\frac{b}{a}$
 - 3) $\frac{a}{b}$
 - 4) $\frac{1}{\text{slope}}$
10. The following are some statements about adsorption of solutes from the solutions.
 - A) Increase in the surface area of the adsorbent increases the extent of adsorption.
 - B) Increase in temperature decreases the extent of adsorption.
 - C) The extent of adsorption (x/m) is related to the molar concentration of solution (c) and is given by $x/m = k \cdot c^{1/n}$
 The correct combination is
 - 1) only A and C
 - 2) only B and C
 - 3) only A and B
 - 4) all are correct

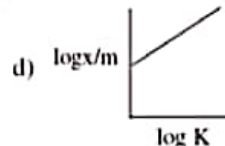
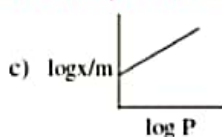
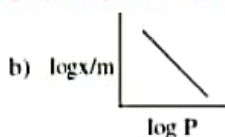
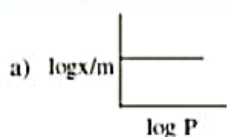
11. The following are some statements about physical adsorption
 A) Involves the weak Vanderwaals interaction between the adsorbate and adsorbent.
 B) Involves the chemical interactions between the adsorbent and adsorbate
 C) is irreversible in nature
 D) It depends upon both nature of adsorbent and adsorbate
 The correct combination is
 1) all are correct 2) only A and B 3) only A and C 4) only A
12. Which one of the following characteristics is not correct for physical adsorption?
 1) Adsorption increases with increases in temperature
 2) Adsorption is spontaneous
 3) Both enthalpy and entropy of adsorption are -ve
 4) Activation energy is slow
13. Amount of gas adsorbed per gram of adsorbent increases with pressure but after certain limit is reached, adsorption becomes constant. It is when
 1) multi layers are formed 2) desorption takes place
 3) temperature is increased 4) absorption also starts
14. Favourable conditions for physical adsorption are
 1) low T, high P 2) high T, high P 3) low T, low P 4) low T, low P
15. Which one of the following gases will be adsorbed most easily?
 1) N_2 2) H_2 3) O_2 4) CO_2

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The volume of gases NH_3 , CO_2 and H_2 adsorbed by one gram of charcoal at 300K are in order of
 a) $H_2 > CO_2 > NH_3$ b) $NH_3 > H_2 > CO_2$ c) $CO_2 > NH_3 > H_2$ d) $NH_3 > CO_2 > H_2$
2. Arsenous sulphide sol is prepared by passing H_2S through arsenous oxide solution the charge developed on the particles is due to adsorption of
 a) H^+ b) S^{-2} c) OH^- d) O^{-2}
3. In Langmuir's model of adsorption of a gas on a solid surface
 a) The mass of gas striking a given area of surface is propotional to the pressure of the gas
 b) The mass of gas striking a given area of surface is independent of the pressure of the gas
 c) The rate of dissociation of adsorbed molecules from the surface does not depend on the surface covered
 d) The adsorption at a single site on the surface may involve multiple molecules at the same time
4. According to adsorption theory of catalysts, the speed of the reaction increases because
 a) The concentration of the reactant molecules at the active centres of the catalyst becomes high due to adsorption
 b) In the process of adsorption, the activation energy of the molecules becomes large
 c) Adsorption produces heat which increases the speed of reaction
 d) Adsoption lowers the activation energy of the reaction

5. Which of the following graph represents Freundlich adsorption isotherm



6. The phenomenon of adsorption arises due to unbalanced

- a) valence force existing on the surface of solids
- b) chemical forces of atoms in the molecules
- c) coulombic forces between electrons and nucleus
- d) electrostatic repulsions between the nuclei

7. Which of the following gases is adsorbed easily and more on activated charcoal

- a) $\text{CO}_2(T_c = 304\text{K})$
- b) $\text{SO}_2(T_c = 430\text{K})$
- c) $\text{H}_2(T_c = 23\text{K})$
- d) all gases undergo adsorption to the same extent

8. According to Freundlich adsorption isotherm, at high pressure, value of x/m is

- a) directly proportional to pressure
- b) inversely proportional to pressure
- c) directly proportional to square of pressure
- d) independent of pressure

9. Adsorption isotherm of gases on solids gives the relation between

- a) volume of adsorbent and temperature
- b) amount of adsorbent per unit weight of adsorbate and pressure
- c) amount of adsorbate per unit weight of adsorbent and equilibrium pressure
- d) volume of adsorbate and pressure

10. Following are the events taking place to explain adsorption theory

I : Desorption

II : Diffusion of the reactants along the surface

III : adsorption of the reactants

IV : formation of the activated surface complex

These events are taking place in the following order

- a) I, II, III, IV
- b) II, III, IV, I
- c) III, IV, I, II
- d) IV, III, II, I

11. Which can adsorb larger volume of hydrogen gas?

- a) Colloidal solution of palladium
- b) Finely divided nickel
- c) Finely divided platinum
- d) Colloidal $\text{Fe}(\text{OH})_3$

More than One correct answer Type Questions

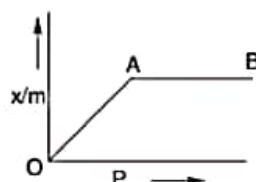
12. Which of the following are correct?

- a) silica gel adsorbs H_2O
- b) CaCl_2 (anhy) adsorbs H_2O
- c) Gas masks work on the principle of selective adsorption
- d) Zeolites are shape selective catalyst and water softners

13. Which of the following give linear plots?

- a) $\log \frac{x}{m}$ versus $\log C$ b) $\log \frac{x}{m}$ versus $\frac{1}{p}$
 c) $\frac{m}{x}$ versus $\frac{1}{p}$ d) $P/\left(\frac{x}{m}\right)$ versus P

14. In the following isotherm (Fig)



- a) $\frac{x}{m} \propto P^0$ When point B is reached b) Desorption may start along AB
 c) $x/m \propto P^{1/n}$ along OA d) $\frac{x}{m} \propto P$ when point B is reached
15. Which statements are correct?
- a) A gas with greater critical temperature gets more adsorbed
 b) The type of adsorption can be determined from the adsorption isobars
 c) Chemisorption is reversible in nature
 d) Variation of extent of adsorption with temperature tells about the type of adsorption
16. Which of the following statements are correct?
- a) Adsorption always leads to a decrease in enthalpy and entropy of the system.
 b) Adsorption arises due to unsaturation in the enthalpy of valence forces of atoms or molecules on the surface.
 c) Adsorption increases with rise in temperature
 d) Adsorption decreases the surface energy.

Linked Comprehension Type Questions

Passage-I :

Adsorption is the tendency of accumulation of molecular species at the surface of solid or liquid. Depending upon the nature of bonds of forces of attraction between adsorbate and adsorbent it is classified between physisorption and chemisorption.

17. Which of the following gas molecules have maximum value of enthalpy of physisorption?
 a) C_2H_6 b) Ne c) H_2 d) H_2O
18. Which of the following gases is adsorbed most by activated charcoal?
 a) CO_2 b) N_2 c) Ar d) C_2H_6
19. Which of the following characteristics is not correct for physical adsorption?
 a) Adsorption is spontaneous b) ΔH and ΔS are negative
 c) It is reversible in nature d) Degree of adsorption increases with temperature

Passage-II :

Langmuir adsorption isotherm is obeyed by the adsorption where the adsorbate forms only a unimolecular adsorbed layer. The mathematical relation of Langmuir adsorption isotherm is:

$$\frac{x}{m} = \frac{aP}{1 + bP}$$

20. For adsorption of a gas on a solid, the plot of $\log\left(\frac{x}{m}\right)$ vs $\log P$ is linear with slope equal to (n being whole number)

a) K b) $\log K$ c) n d) $\frac{1}{n}$

21. The equation of Langmuir adsorption isotherm under high pressure is:

a) $\frac{x}{m} = \frac{a}{b}$ b) $\frac{x}{m} = aP$ c) $\frac{x}{m} = \frac{1}{aP}$ d) $\frac{x}{m} = \frac{b}{a}$

Matrix Matching Type Questions

22. **Column-I**

A) Chemisorption
B) Physisorption
C) Desorption of a solute from liquid
D) Adsorption of a solute on a liquid

Column-II

p) Not very specific and decreases with temp.
q) Specific and increases with temp.
r) Increases the surface tension of the liquid surface
s) Adsorption is due to stronger interaction or bond formation

23. **Column-I**

A) Physisorption
B) Chemisorption
C) Activated adsorption
D) Desorption

Column-II

p) Multimolecular
q) High heat of activation
r) High temperature required
s) Low pressure required

Integer Type Questions

24. At STP the volume of nitrogen gas required to cover a sample of silica gel, assuming Langmuir monolayer adsorption, is found to be $1.33\text{cm}^3\text{g}^{-1}$ of the gel. The area occupied by a nitrogen molecule is 0.14nm^2 . What is the surface area per gram of silica gel (in m^2) ?
25. In an adsorption experiment, a graph between $\log(x/m)$ versus $\log P$ was found to be linear with a slope of 45° . The intercept on the y-axis was found to be 0.301. Calculate the amount of the gas adsorbed per gram of charcoal under a pressure of 3.0 atm.

EXERCISE-II

(Colloids)

LEVEL-I (MAIN)**Straight Objective Type Questions**

- Particles of which of the following do not pass through ultra filter paper
 - colloids only
 - true solutions
 - suspensions only
 - colloids and suspension
- The tyndall effect in colloidal solutions is due to
 - scattering of light
 - reflection of light
 - absorption of light
 - charge of particles

3. Which of the following is a homogeneous system?
 1) suspension 2) colloid solution 3) true solution 4) starch solution
4. When the dispersed phase has a greater affinity for the dispersion medium, the colloids are termed as
 1) lyophilic 2) lyophobic 3) hydrophobic 4) emulsion
5. The dispersed phase, dispersion media, and the nature of colloidal solution of gold sol respectively are
 1) Solid, solid, lyophobic 2) Liquid, liquid, lyophobic
 3) Solid, liquid, lyophobic 4) Solid, liquid, lyophilic
6. Gold numbers of protective colloids A, B, C and D are 0.50, 0.01, 0.10, and 0.005 respectively. The correct order of their protective power is
 1) $C < B < D < A$ 2) $A < C < B < D$ 3) $A < B < C < D$ 4) $D < A < C < B$
7. Which of the following is a non electrolytic colloidal sol?
 1) Starch 2) AgCl sol 3) Arsenic sulphide sol 4) Sb_2S_3 sol
8. In Faraday-Tyndall effect the colloiddally suspended particles
 1) trace out the path of strong beam of light 2) coagulate
 3) show electrophoresis 4) show brownian movement
9. Which one of the following constitutes irreversible colloidal system with water as dispersion medium?
 1) Clay 2) Platinum 3) $Fe(OH)_3$ 4) All of three
10. Insecticide sprays are example of
 1) Liquid in gas 2) Gas in liquid 3) Gas in solid 4) Solid in liquid
11. Bredig's arc method is used for the preparation of colloidal solution of
 1) Organic compounds 2) metals like silver, gold etc
 3) two liquids 4) inorganic compounds
12. There is no scum formation when hard water is being used. The washing powder can be
 1) $C_{17}H_{35}COONa$ 2) $R-\text{C}_6\text{H}_4-\text{SO}_3\text{Na}$ 3) both 4) none
13. Non-ionic surfactants are
 1) $R-\text{C}_6\text{H}_4-\text{SO}_3\text{Na}$ 2) $C_{17}H_{35}COONa$
 3) $C_nH_{2n+1}(OCH_2CH_2)_mOH$ 4) All
14. Micelles are
 1) Ideal solution 2) Associated colloids 3) Adsorbed surfaces 4) Absorbent solutes
15. A sol is prepared by addition of excess of $AgNO_3$ solution in KI solution. The charge likely to develop on colloidal particles is
 1) Negative 2) Positive 3) No charge 4) Both charges
16. The gold number of four protective colloids A, B, C and D are 0.03, 0.003, 10 and 30 respectively. Protective power of these colloids will be in the order:
 1) $B > A > C > D$ 2) $A > B > C > D$ 3) $C > B > D > A$ 4) $D > A > C > B$
17. Cottrell precipitator works on the principle of
 1) distribution law 2) addition of electrolyte
 3) neutralisation of charge on colloids 4) Le Chatelier's principle

18. What is the emulsifier in milk?

1) Albumin

2) Soap

3) Gelatin

4) Caesin

Numerical Value Type Questions

19. For the coagulation of 50ml of ferric hydroxide sol 10ml of 0.5M KCl is required. What is the coagulation value of KCl

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. Which one of the following has maximum value of flocculation power

a) Pb^{2+}

b) Pb^{4+}

c) Sr^{2+}

d) Na^+

2. Which of the following electrolytes will be most effective ion in the coagulation of gold sol

a) $NaNO_3$

b) $K_4[Fe(CN)_6]$

c) Na_3PO_4

d) $MgCl_2$

3. The coagulating power of an electrolyte for Arsenous sulphide sol decreases in the order

a) $Na^+ > Al^{3+} > Ba^{2+}$

b) $PO_4^{3-} > SO_4^{2-} > Cl^-$

c) $Cl^- > SO_4^{2-} > PO_4^{3-}$

d) $Al^{3+} > Ba^{2+} > Na^+$

4. The coagulation of 200 ml of a positive colloid took place when 0.73g HCl was added to it without changing the volume much. The flocculation value of HCl for the colloid is

a) 0.365

b) 36.5

c) 100

d) 200

5. Blue colour of the sky is due to

a) Absorption of light by dust particle

b) Scattering of light by dust particle

c) Reflection of light by dust particle

d) Presence of clouds

6. The volume of a colloidal particle V_c as compared to the volume of a solute particle in a true solution V_s could be

a) $\frac{V_c}{V_s} = 10^3$

b) $\frac{V_c}{V_s} = 10^{-3}$

c) $\frac{V_c}{V_s} = 10^{23}$

d) $\frac{V_c}{V_s} = 1$

7. The apparatus used to coagulate carbon particles from smoke is called

a) Cottrell smoker

b) Cottrell absorber

c) Cottrell precipitator

d) None

8. When freshly precipitated $Fe(OH)_3$ is boiled with water in the presence of few drops of dil HCl, a hydrated ferric oxide sol is obtained. This method is termed

a) Dialysis

b) Peptization

c) ultrafiltration

d) Electro dispersion

9. Which of the following forms micelles in solutions of higher concentrations

a) Sodium carbonate

b) Sodium bicarbonate

c) Potassium acetate

d) Sodium stearate

10. A substance which forms micelles in solutions contains

a) carboxylic group

b) alkyl group

c) water insoluble long hydrocarbon groups and water soluble polar groups

d) water soluble hydrocarbon group and water insoluble polar group

11. A lyophobic sol is at its isoelectric point then it is:

a) negatively charged

b) positively charged

c) not charged

d) none of these

12. During micelle formation:

- a) $\Delta H = +ve$; $\Delta S = +ve$ b) $\Delta H = -ve$, $\Delta S = +ve$
 c) $\Delta H = -ve$, $\Delta S = +ve$ d) $\Delta H = +ve$, $\Delta S = -ve$

13. Gel on standing excludes small amounts of liquid. This phenomenon is known as

- a) syneresis b) efflorescence c) adsorption d) thixotropy

14. A micelle formed during the cleansing action by soap is

- a) a discrete particle of soap b) an aggregated particles of soap and dirt
 c) a discrete particle of dust d) an aggregated particle of dust and water

More than One correct answer Type Questions

15. Anionic surfactants are

- a) $C_{15}H_{31}COONa$ b) $R-\text{C}_6\text{H}_4-\text{SO}_3\text{Na}$ c) $C_{18}H_{37}NH_3Cl$ d) All

16. Cationic surfactants are

- a) $C_{17}H_{13}COONa$ b) $R-\text{C}_6\text{H}_4-\text{SO}_3\text{Na}$
 c) $C_{16}H_{33}-\text{C}_6\text{H}_4-\text{N}^+\text{Cl}^-$ d) $C_{16}H_{33}N(CH_3)_3Cl$

17. Which of the following is represented by sols

- a) Adsorption b) Tyndal effect c) Flocculation d) Paramagnetism

18. Select the properties which are for lyophilic colloidal sols

- a) Viscosity same as that of the medium b) Extensive hydration takes place
 c) Particles migrate either towards cathode or anode in an electric field
 d) Particles cannot be detected even under ultramicroscopes

19. Colloidal gold can be prepared by

- a) reduction of AuCl_3 b) Bredig's arc method
 c) hydrolysis d) Peptization

20. Coagulation or de-emulsification can be done by which of the methods given below.

- a) by addition of a substance which would destroy the emulsifier
 b) by addition of an electrolyte which would destroy the charge
 c) by heating, freezing and centrifuging
 d) by electro phoresis

21. Bleeding of blood is stopped due to

- a) the coagulation of blood by applying FeCl_3 and blood vessel is sealed
 b) the coagulation of blood by applying Alum and blood vessel is sealed
 c) blood combines with the FeCl_3
 d) blood combines with Alum

22. At CMC, the surfactant molecules undergo:

- a) Ionization b) aggregation
 c) micelle formation d) reduction in the surface tension of water

23. Which among the following are lyophilic in nature?

- a) Starch b) Gum c) Gelatin d) Gold

Linked Comprehension Type QuestionsPassage :

Colloidal solution is a heterogeneous solution which contains particle of intermediate size, i.e., (diameter between 1 and 1000 nm) colloidal is not a substance but it is a state of a substance which depends upon the molecular size. Colloidal solutions are intermediate between true solutions and suspensions.

24. The size of the colloidal particles lies in the range
 a) 10nm – 1000nm b) 10m μ – 1000m μ c) 1nm – 1000nm d) 10⁻⁵ cm – 10⁻⁷ cm
25. The colloidal solution of a solid as the dispersed phase and a gas as the dispersion medium is called
 a) Sol b) Solid foam c) Aerosol d) Gel

Matrix Matching Type Questions26. **Column-I**

- A) Brownian motion
 B) Tyndall effect
 C) Particle Size 10⁻⁹ – 10⁻⁸ m
 D) Completely transparent

Column-II

- p) Colloidal solution
 q) True solution
 r) Suspension
 s) Emulsion
 t) Scattering of light

27. **Column-I**

- A) Bredig's arc method
 B) Electro dialysis
 C) Ultra filtration
 D) Peptization

Column-II

- p) Movement of ions towards oppositely charged electrodes
 q) Preparation of metal sols
 r) Purification of sols
 s) Preparation of colloidal solution
 t) The precipitate adsorbs one of the ions of the electrolyte on its surface.

28. **Column-I**

- A) Mist
 B) Rain Cloud
 C) Jellies
 D) Whipped cream

Column-II

- p) Gel
 q) Foam
 r) Aerosol
 s) Solid or liquid dispersed in air

Integer Type Questions

29. The no of moles of lead nitrate needed to coagulate 2 moles of colloidal [AgI] I⁻ is ____
30. In an experiment addition of 10ml of 0.05 M BaCl₂ to 20ml of As₂S₃ sol causes coagulation in 1 hr. If the coagulation value of Ba²⁺ ion is x²/3 then x = ____?
31. When 1.0 × 10⁻⁵ mg of a protective colloid is added to 20 ml of standard gold sol, the precipitation of gold sol was prevented by adding 5 ml of 10% solution of NaCl. If the gold number of protective colloid is 10⁻⁸ then x is
32. The particle size of suspension should be greater than 10⁵Å⁰, x is ____
33. 548 ml of 0.5 M HCl is shaken with 0.5 g of activated charcoal and filtered. The concentration of the filtrate is reduced to 0.4 M. The amount of adsorption (x/m) is:
34. In an experiment, addition of 5.0 ml, of 0.005 M BaCl₂ to 10.0 ml of arsenic sulphite sol just causes the complete coagulation in 100 min. The flocculating value of the effective ion is :

EXERCISE-III

(Catalysis)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Catalyst increases the rate by
 - Decreasing E_a
 - Increasing E_a
 - Decreasing Pressure
 - Increasing entropy
- Which of the following catalyst is used during the hydrogenation of oil?
 - Fe
 - Ni
 - Pt
 - Mo
- Which of the following is present at the time of cracking of hydrocarbons?
 - Copper
 - Zeolite
 - Nickel
 - Molybdenum
- Which is not the correct statement for a catalyst?
 - It does not alter E_a
 - The surface of a catalyst adsorbs reactants
 - Catalyst may form intermediates with reactants
 - Action of enzyme catalyst is always specific
- Match Column-A (Catalyst) with Column-B (process)

A	B
1) Fe	I) Cracking of hydrocarbon
2) Pt	II) Haber process
3) Zeolites	III) Automobile converter
1) I-I; 2-II; 3-III	3) I-II; 2-III; 3-I
2) I-III; 2-II; 3-I	4) I-III; 2-I; 3-II
- Catalyst used in polymerization of ethene is
 - $TiCl_4$ and AlR_3
 - Fe, Co
 - H_3PO_4
 - Zeolites
- Energy of activation of forward and backward reaction are equal in cases (numerical values) where
 - $\Delta H = 0$
 - No catalyst present
 - $\Delta S = 0$
 - Stoichiometry is the mechanism
- What is the role of a catalyst in a catalyzed reaction?
 - Lowers the activation energy
 - Increases the activation energy
 - Affects the free energy change
 - Affects the enthalpy change
- The role of a catalyst in a reversible reaction is to
 - Increase the rate of forward reaction only
 - Decrease the rate of backward reaction only
 - Alter the equilibrium constant of the reaction
 - Allow the equilibrium to be achieved quickly
- Which of the following catalyzes the conversion of glucose into ethanol?
 - Zymase
 - Invertase
 - Maltase
 - Diastase
- Which of the following affects the adsorption of gas on the surface of solid?
 - Critical temperature
 - Pressure of gas
 - Temperature
 - All of these
- An inhibitor is essentially:
 - an autocatalyst
 - a negative catalyst
 - a catalyst promoter
 - a homogeneous catalyst

13. An example for homogeneous catalysis is:

- | | |
|-------------------------|-------------------------|
| 1) Contact process | 2) Haber's process |
| 3) Lead chamber process | 4) Hydrogenation of oil |

14. For chemisorption, which of the following is wrong?

- | | |
|--|----------------------------------|
| 1) irreversible | 2) it requires activation energy |
| 3) it forms multi molecular layer on adsorbate | 4) surface compounds are formed |

Numerical Value Type Questions

15. In presence of catalyst, the activation energy is lowered by 3 k cal at 27°C. Hence, the rate of reaction will increase by:

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- Which of the following process does not occur at the interface of phases?

a) Crystallisation	b) Heterogeneous catalysis
c) Homogeneous catalysis	d) Corrosion
- Hydrolysis of protein in stomach and in intestine takes place due to action of enzyme:

a) pepsin in stomach, trypsin in intestine	b) trypsin in stomach, pepsin in intestine
c) both (a) and (b)	d) none of the above
- Which of the following not property of catalyst?

a) A catalyst remains unchanged chemically at the end of chemical reaction
b) A catalyst takes part in a chemical reaction
c) All kinds of catalysis undergo catalytic poisoning
d) A catalyst physically changes at the end of reaction
- The rate of a certain biochemical reaction catalysed by an enzyme in human body is 104 times faster than when it carried out in the laboratory. The activation energy of this reaction:

a) is zero	b) is different in two cases
c) is the same in both the cases	d) none of the above
- Air can oxidize sodium sulphite in aqueous solution but cannot do so in the case of sodium arsenite. If however, air is passed through a solution containing both sodium sulphite and sodium arsenite then both are oxidized. This is an example of

a) positive catalysis	b) negative catalysis
c) induced catalysis	d) auto-catalysis
- In the reversible reaction, a catalyst is the substance which

a) Increases the rate of the forward reaction
b) Decreases the value of enthalpy change in the reaction
c) Reduces the time required for reaching the equilibrium state in the reaction
d) Decreases the rate of the reverse reaction
- The components of Zigler-Natta catalyst are

a) $\text{TiCl}_3 + \text{Al}(\text{C}_2\text{H}_5)_3$	b) $\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$	c) $\text{Ti}(\text{C}_2\text{H}_5)_3 + \text{AlCl}_3$	d) $\text{Ti}(\text{C}_2\text{H}_5)_4 + \text{AlCl}_3$
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- The decomposition of H_2O_2 , is slowed down by the addition of small amount of acetamide. Acetamide acts as :

a) promoter	b) stopper	c) inhibitor	d) poison
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9. Catalyst can alter:

- a) The amount of product
- b) heat of reaction
- c) rate constant
- d) equilibrium constant

More than One correct answer Type Questions

10. Which of the following statements are correct about solid catalyst?

- a) Same reactants may give different products by using different catalysts
- b) Catalyst does not change ΔH of reaction.
- c) Catalyst is required in large quantities to catalyse reaction
- d) Catalytic activity of a solid catalyst does not depend upon the strength of chemisorption

11. In a reaction, catalyst changes ____

- a) physically
- b) qualitatively
- c) chemically
- d) quantitatively

12. Which act(s) as negative catalyst?

- a) Lead tetraethyl as antiknock compound
- b) Glycerol in decomposition of H_2O_2
- c) Ethanol in oxidation of chloroform
- d) None of the above

13. Which of the following statements is/are true?

- a) Iron is used as a catalyst in the hydrogenation of oils
- b) V_2O_5 is used as a catalyst in the oxidation of SO_2 to SO_3
- c) Haber's process requires iron as a catalyst
- d) Thermite process does not involve any catalyst

14. Which of the following statements is/are true for heterogeneous catalysis?

- a) The energy of activation does not change
- b) The catalyst combines with the reactant to form a compound
- c) The reactant molecules are adsorbed on the surface of the catalyst
- d) The catalyst lowers the energy of activation

15. Which of the following is/are not affected by the catalyst?

- a) Heat of reaction
- b) Equilibrium constant
- c) Amount of product
- d) Rate constant of the reaction

16. Which of the following are correct about the catalyst?

- a) Physical adsorption is of multimolecular layer
- b) Degree of chemical adsorption increases with increase in temperature
- c) Adsorption increases the surface energy
- d) Sometimes solvent is adsorbed in preference to solution

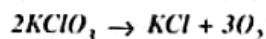
17. Select the correct statements about enzymes:

- a) Enzymes are biological catalysts found in organisms
- b) All enzymes are proteins
- c) Enzymes can catalyse any reaction
- d) Enzymes activity is optimum at 27°C

Linked Comprehension Type Questions

Passage :

Catalysis are not consumed in the reaction, hence very small non-stoichiometric quantities are generally required. For example when potassium chlorate heated strongly decomposes slowly giving dioxygen, the decomposition occurs in the temperature range of 653-873K.



18. The oxidation of oxalic acid by acidified KMnO_4 becomes faster with the passage of time due to:
- Presence of H^+ ions
 - autocatalysis by Mn^{2+}
 - Presence of MnO_4^- ions
 - Presence of K^+ ions
19. In case of autocatalysis:
- product acts as catalyst
 - reactant acts as catalyst
 - solvent acts as catalyst
 - heat produced in the reaction catalysis

Matrix Matching Type Questions

20. Column-I

- Quinoline
- Acetamide
- Zeolite
- Nickel

Column-II

- Inhibitor for decomposition of H_2O_2
- Catalyst
- Removes hardness of water
- Poison for Pd in Lindlar's catalyst

21. Column-I

- Nickel
- V_2O_5
- Diastase
- Adsorption

Column-II

- Conversion of SO_2 into SO_3
- Conversion of starch into sugar
- Accumulation of molecules at the surface of a solid or liquid
- Hydrogenation of vegetable oils

KEY SHEET (PRACTICE SHEET)

EXERCISE-I

LEVEL-I

- 2
- 3
- 3
- 2
- 3
- 2
- 2
- 2
- 2
- 4
- 4
- 1
- 1
- 1
- 4

LEVEL-II

- d
- b
- a
- d
- c
- a
- b
- d
- c
- b
- a
- acd
- acd
- abc
- abd
- abd
- d
- a
- d
- d
- a
- A-qt; B-p; C-r; D-ps
- A-p; B-qr; C-qr; D-s
- 5
- 6

EXERCISE-II

LEVEL-I

- 4
- 1
- 3
- 1
- 3
- 2
- 1
- 1
- 4
- 1
- 2
- 2
- 3
- 2
- 2
- 1
- 3
- 4
- 100

OBJECTIVE CHEMISTRY IID**SURFACE CHEMISTRY****LEVEL-II**

- 1) b 2) d 3) d 4) c 5) b 6) a 7) c 8) b
9) d 10) c 11) d 12) a 13) a 14) b 15) ab 16) cd
17) abc 18) bd 19) ab 20) abcd 21) ab 22) bcd 23) abc 24) c
25) c 26) A-pqrs; B-prst; C-ps; D-q 27) A-qs; B-pr; C-r; D-st
28) A-rs; B-rs; C-p; D-q 29) 1 30) 5 31) 6 32) 3 33) 4
34) 2

EXERCISE-III**LEVEL-I**

- 1) 1 2) 2 3) 2 4) 1 5) 3 6) 1 7) 1 8) 1
9) 4 10) 1 11) 4 12) 2 13) 3 14) 3 15) 148

LEVEL-II

- 1) c 2) a 3) d 4) b 5) c 6) c 7) b 8) c
9) c 10) ab 11) ab 12) abc 13) bcd 14) bcd 15) abc 16) abd
17) abd 18) b 19) a 20) A-s; B-p; C-qr; D-q
21) A-s; B-p; C-q; D-r